

ECR-2 SIMULATION REVIEW

LLX-RND-2026-0003 | Frozen Run Audit

Run #948 26/08/2026, 08:18:08 am IST

Run review and engineering disposition

This report reviews the persisted ECR-2 Simulator snapshot for Run #948. It does not rerun the engine, alter the database, or reconstruct missing results. The run was created from Revision 8 on 26/08/2026, 08:18:08 am IST.

DISPOSITION: NOT CALCULABLE. The run failed closed before an accepted physical BVP, ideal-stage cascade, theoretical stage count, or RRBO recovery was available. This is a calculation/dependency block, not a finding that the process is infeasible.

Run identity

Item	Stored value
Run ID	#948
Revision	8
Created	Wed Aug 26 2026 02:48:08 GMT+0000 (Coordinated Universal Time) UTC
User-facing time	26/08/2026, 08:18:08 am IST
Calculation type	ecr_simulator
Engine	llx-ecr-simulator v2.1.0
Calculation class	Preliminary Screening
Persisted run status	ERROR
Calculation status	counter_current_bvp_not_accepted
Reporting status	NOT_CALCULABLE

1. Frozen input basis

Values below are read from input_snapshot; blanks remain reported as not recorded.

Item	Stored value
Operating / extraction temperature	50 deg C / 323.15 K
RRBO feed	3448 kg/h
NMP feed	4024 kg/h
Solvent-to-oil ratio	1.167053
NMP purity	0.995
Phase configuration	nmp_continuous_rrbo_dispersed
Raffinate aromatic target	10 mol%; hydrocarbon-only physical outlet basis
Requested theoretical stages	1
User column height	14 m

Item	Stored value
Column diameter input	NOT RECORDED m
Stage 8 system acceptance	ACCEPTED

RRBO component	Stored mass fraction	Approx. feed flow
Saturates	0.85	2930.8 kg/h
Mono-aromatics	0.05	172.4 kg/h
Di-aromatics	0.05	172.4 kg/h
Poly-aromatics	0.05	172.4 kg/h
Total RRBO	1	3448 kg/h

2. Gate-by-gate review

Each gate is classified from the frozen result snapshot.

Gate / surface	Stored disposition	Review finding
Thermodynamic coordinate mapping	BLOCKED	Coto surrogate mole coordinates are not a validated physical RRBO pseudo-component balance.
Ideal-stage cascade	THERMODYNAMIC SURROGATE-TO-PHYSICAL MAPPING NOT CLOSED	nT is null; zero trials; no selected stage count; no recovery reconciliation.
Physical height sizing	NOT_CALCULABLE	Initial 0.0100 m trial rejected because K&H 1995 holdup is physically invalid.
Counter-current BVP	BLOCKED	No accepted outlets, residual vector, axial profile, or compartments persisted.
Independent 30% compartments	BLOCKED_BY_IDEAL_STAGE_CASCADE	Correctly not run before a same-specification ideal-stage result existed.
Transfer output	LOCAL_PRELIMINARY_BLOCKED	Governed transfer values unavailable; release status not eligible.
Optimizer	NOT IMPLEMENTED - downstream of BVP	Downstream of an accepted BVP and therefore not executable.

Primary blocker: THERMODYNAMIC SURROGATE-TO-PHYSICAL MAPPING NOT CLOSED. The current model must not use either surrogate molecular weights or physical RRBO pseudo-component molecular weights as a validated bridge into the Coto NRTL coordinate. No ideal-stage NRTL recovery is therefore calculated.

- The historical 53.6715% recovery is explicitly labelled an unvalidated historical sensitivity and is not a Run #948 result.
- The absence of a theoretical stage count is a calculation block; it must not be translated into process infeasibility.

3. Ideal-stage and BVP result evidence

Item	Stored value
Ideal-stage status	THERMODYNAMIC SURROGATE-TO-PHYSICAL MAPPING NOT CLOSED
Theoretical stage count, nT	NOT RECORDED
Ideal-stage trial count	0
Selected ideal-stage trial	NONE
Independent recovery reconciliation	NOT AVAILABLE
Ideal-stage diagnostic count	4
BVP status	BLOCKED

Item	Stored value
BVP iterations	0
BVP outlets	NONE
BVP failure	Required Active Extraction Height = NOT_CALCULATED because The initial 0.0100 m physical-height BVP is not accepted: K&H 1995 holdup is unusable: physically_invalid.
Required active height	NOT RECORDED m
Required height status	NOT_CALCULATED
Recovery result	NOT CALCULATED

Persisted ideal-stage diagnostics:

- The Coto NRTL parameters are fitted to n-dodecane/1,4-xylene/1-methylnaphthalene/pyrene/NMP mole coordinates, while the project RRBO physical pseudo-components have different molecular weights.
- A one-to-one RRBO-class !" Coto-surrogate screening analogy does not define a physical-pseudo-mole !" surrogate-mole mapping or a valid NRTL material-balance bridge.
- THERMODYNAMIC SURROGATE-TO-PHYSICAL MAPPING NOT CLOSED: neither the surrogate-MW coordinate nor the physical-pseudo-component-MW coordinate can be supplied to the current Coto NRTL model as a validated physical ECR-2 material balance.
- No ideal-stage NRTL flash, physical product quality, RRBO recovery, or theoretical-stage result is calculated until the mapping is governed. Previously calculated 53.6715% recovery is an unvalidated historical sensitivity, not a validated physical RRBO recovery.

The frozen snapshot contains no physical raffinate or extract outlet flows from which to calculate product quality or NMP-free RRBO recovery.

4. Thermodynamic and physical-basis limitations

Item	Stored value
Operating temperature classification	EXTRAPOLATED
Selected temperature	50 deg C / 323.15 K
Nearest admitted evidence	298.15 K
Distance outside admitted range	25 K
Selected-temperature applicability	EXTRAPOLATED_PRELIMINARY
NRTL calibration reproduction	FAIL
Tie-lines within gate	2 / 13
Maximum calibration deviation	0.092
Independent multi-temperature validation	INSUFFICIENT_EVIDENCE
Optimizer-forward thermodynamic readiness	NOT_READY
d32 status	DIRECT_TURBULENCE_D32_PRELIMINARY - PRELIMINARY_ENGINEERING / NOT YET PILOT_VALIDATED
d32 nominal	0.976126 mm
Holdup status	physically_invalid
Holdup raw value	10.418884

At 50 deg C the run-level LLE flash is classified preliminary temperature extrapolation from the admitted 298.15 K evidence point. The active NRTL form is calibration-temperature-only, and its reproduction check fails the 3·u(x) gate (2 of 13 tie-lines within gate; maximum absolute deviation 0.092). This is recorded as a limitation, not hidden as a successful validation.

5. Persisted warnings and final disposition

Code	Stored warning
INPUT_WARNING	operatingTemperatureC: NRTL model: temperature 50 degC is outside the calibrated range - NRTL equilibrium targets will be extrapolated (Pending Validation).
INTERFACIAL_TENSION_TEMPERATURE_DEPENDENCE_NOT_MODELLED	sigma uses the unchanged 70 degC source anchor at the selected 50 degC operating condition under the governed preliminary constant-over-range basis [25, 100] degC. No sigma(T) correlation has been applied.
ECR2_PROCESS_SIZING_BASIS_PRELIMINARY	ECR-2 generated its diameter trials and $C_{F+R} = 7FV_0 \cdot \sqrt{V \cdot E} - \alpha \cdot y$ process-model basis. This evaluation envelope is not a hydraulic/flooding criterion and does not select a final diameter.
PHYSICAL_MW_ASSUMED	One or more physical/project RRBO molecular weights are Assumed. They remain active engineering inputs for the pending physical mass-transfer architecture and do not alter Coto/NRTL coordinates.
NMP_PURITY_PHYSICAL_ONLY	NMP purity 0.9950 is retained as a physical plant-boundary input. The C2 thermodynamic solvent coordinate is canonical fresh NMP; characterize impurities before any physical mass-transfer architecture is enabled.
EPD_ASSUMED_VALUE	Tabular point at 25 degC is ASSUMED (Virgin Base oil - SN300 (Assumed - provisional; replace with controlled vendor or measured data before release-grade use)) - replace with measured/vendor data before release-grade use.
EPD_ASSUMED_VALUE	Tabular point at 30 degC is ASSUMED (Virgin Base oil - SN300 (Assumed - provisional; replace with controlled vendor or measured data before release-grade use)) - replace with measured/vendor data before release-grade use.
EPD_ASSUMED_VALUE	Tabular point at 40 degC is ASSUMED (Virgin Base oil - SN300 (Assumed - provisional; replace with controlled vendor or measured data before release-grade use)) - replace with measured/vendor data before release-grade use.
EPD_ASSUMED_VALUE	Tabular point at 50 degC is ASSUMED (Virgin Base oil - SN300 (Assumed - provisional; replace with controlled vendor or measured data before release-grade use)) - replace with measured/vendor data before release-grade use.
EPD_ASSUMED_VALUE	Tabular point at 60 degC is ASSUMED (Virgin Base oil - SN300 (Assumed - provisional; replace with controlled vendor or measured data before release-grade use)) - replace with measured/vendor data before release-grade use.
EPD_ASSUMED_VALUE	Tabular point at 70 degC is ASSUMED (Virgin Base oil - SN300 (Assumed - provisional; replace with controlled vendor or measured data before release-grade use)) - replace with measured/vendor data before release-grade use.
HOLDUP_PHYSICALLY_INVALID	K&H 1995 holdup: $\phi_{raw} = 10.4189$ ($\phi_{raw} \geq 1$) - the correlation has produced a physically inadmissible prediction at this operating point. Review agitation intensity ($\alpha_B f \cdot E_{ow}$ conditions.
D32_DIRECT_TURBULENCE_PRELIMINARY	DIRECT_TURBULENCE_D32_PRELIMINARY d32 = 0.976 mm (C=0.39; C sensitivity 0.901-1.076 mm). PRELIMINARY_ENGINEERING / NOT YET PILOT_VALIDATED; separate from K&H 1996.
ECR2_HEIGHT_NOT_CALCULABLE	ECR-2 physical height is not calculable: The initial 0.0100 m physical-height BVP is not accepted: K&H 1995 holdup is unusable: physically_invalid.
BVP_NOT_ACCEPTED	Required Active Extraction Height = NOT_CALCULATED because The initial 0.0100 m physical-height BVP is not accepted: K&H 1995 holdup is unusable: physically_invalid.
BVP_MASS_BALANCE_FAILED	Counter-current BVP result is not accepted because its mass-balance check did not pass.

Persisted validation issues:

Code	Stored issue
ISSUE	NRTL model: temperature 50 degC is outside the calibrated range - NRTL equilibrium targets will be extrapolated (Pending Validation).

FINAL ENGINEERING DISPOSITION: Run #948 is NOT CALCULABLE and NOT RELEASE ELIGIBLE. It establishes no valid ECR-2 N_T, no hydrocarbon-only raffinate aromatic result, no NMP-free RRBO recovery, no accepted physical height, and no governed transfer performance.

Recommended next action: close the surrogate-to-physical mapping with direct evidence or a separately governed calibration, then create a new run. The new run must independently pass the ideal-stage recovery reconciliation, physical BVP acceptance, terminal component/total balances, and the 95% NMP-free RRBO recovery requirement before any result can support engineering selection.